

1 2D-DCT Transform

(a) For $A^{4 \times 4}$, $N=4$. Therefore, the DCT matrix T should be:

$$T[i, j] = \begin{cases} \sqrt{\frac{1}{N}} = \frac{1}{2}, & \text{if } i = 0 \\ \sqrt{\frac{2}{N}} \cos\left(\frac{(2j+1)i\pi}{2N}\right) = \frac{1}{\sqrt{2}} \cos\left(\frac{(2j+1)i\pi}{8}\right), & \text{if } 0 < i < N, 0 \leq j < N \end{cases}$$

$$T = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} \cos \frac{\pi}{8} & \frac{1}{2} \cos \frac{3\pi}{8} & \frac{1}{2} \cos \frac{5\pi}{8} & \frac{1}{2} \cos \frac{7\pi}{8} \\ \frac{1}{2} \cos \frac{2\pi}{8} & \frac{1}{2} \cos \frac{6\pi}{8} & \frac{1}{2} \cos \frac{10\pi}{8} & \frac{1}{2} \cos \frac{14\pi}{8} \\ \frac{1}{2} \cos \frac{3\pi}{8} & \frac{1}{2} \cos \frac{9\pi}{8} & \frac{1}{2} \cos \frac{15\pi}{8} & \frac{1}{2} \cos \frac{21\pi}{8} \end{bmatrix} \approx \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.653281 & 0.270598 & -0.270598 & -0.653281 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.270598 & -0.653281 & 0.653281 & -0.270598 \end{bmatrix}$$

As for the DCT transform consequence $D = TAT^T$

$$\text{Therefore, the consequence } D = \begin{bmatrix} 10.0000 & 9.2388 & 0.0000 & -3.8268 \\ 9.2388 & 8.5355 & 0.0000 & -3.5355 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ -3.8268 & -3.5355 & 0.0000 & 1.4645 \end{bmatrix}. \quad \text{(b) Comparing with the}$$

2D-DFT, the 2D-DCT is easier to compute. The DCT only consider the cosine computation. The DFT must consider cosine and sine computation at same time. They will have similar performance on showing the color frequency on images. However, the DCT could save half of computation cost. Therefore, DCT performs better from my perspective.

2 Controllable Video Generation

(a) Cross-subject motion-guided video generation aims to transfer actions from a source video (e.g., a person running) to a different target subject (e.g., a horse or lion), preserving motion semantics while adapting appearance. This approach leverages video signal processing techniques, such as spatiotemporal feature extraction, skeleton/keypoint estimation, and frame-wise motion modeling, enabling temporally coherent video synthesis. The motivation is to explore novel applications in animation, virtual character creation, and behavior simulation, while addressing challenges in motion-appearance disentanglement and realistic video generation (80 Words).

(b).

References

- [1] Chan, C. et al., “Everybody Dance Now”, CVPR 2019.
- [2] Siarohin, A. et al., “First Order Motion Model for Image Animation”, NeurIPS 2019.
- [3] Zhang, X. et al., “Pose-Guided Person Image Generation”, ICCV 2019.